

PES UNIVERSITY, Bangalore
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Department of Computer Science & Engineering
UE24CS341A : Software Engineering

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Lab 3: Architectural Pattern Selection — Written Justification

Problem Statement #64 — Disaster Relief Supply & Volunteer Coordinator

Reason 1 — Independent Scaling & Fault Isolation for Critical Functions

The system has different functions such as requisition prioritization (FR-001), volunteer skill matching and dispatch (FR-002, FR-003), and vehicle tracking with ETA alerts (FR-004, FR-005). During a disaster, the workload may not be the same for all these functions. For example, many shelters may send supply requests while only a few vehicles are being tracked.

By using microservices, each service can be scaled independently based on its workload. Also, if one service such as the Volunteer Coordination Service becomes slow or fails, other services like Vehicle Tracking and Notification can continue to work. This is important in a disaster management system because critical activities such as water or medical supply dispatch should not be stopped due to failure of an unrelated service.

Reason 2 — Independent Deployment Supports Offline-Capable, Low-Bandwidth Clients

The System needs to work properly even when the network connection is slow or unavailable. NFR-001 requires the dashboard to support low bandwidth and offline GIS caching, while NFR-002 requires Disaster Managers and Relief Volunteers to continue working offline without losing data.

A separate Offline Sync and Cache Service can handle these requirements. This service can be updated, optimized, or scaled independently without affecting other services such as Requisition, Volunteer Coordination, or Vehicle Tracking. This makes the system more suitable for disaster situations where network connectivity may be unreliable.

Security Advantage

Each microservice can have its own access control and database. For example, only the Requisition and Prioritization Service needs access to the external Warehouse System, while the Volunteer Coordination Service handles volunteer skill profiles and related personal information.

Therefore, if one service is compromised, the damage can be limited to that particular service and its data. This reduces the overall security risk compared to a monolithic system where different functions may directly access a common database.

Performance Benefit

FR-005 requires an ETA- exceeded alert to reach the Disaster Manager within 2 minutes. The Notification and Alert Service can be deployed separately and kept lightweight so that it can process and send alerts quickly.

Since it is independent, the notification service will not be heavily affected by other workloads such as GIS processing or requisition matching. It can also be scaled separately when the number of alerts increases. This helps maintain low and predictable alert response time, especially during large-scale disasters.